

Выпуклые функции: субдифференциал и двойственность

Homework

1 сентября 2026 года

1. (Sierpiński 1920) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be Lebesgue measurable and midpoint convex. Prove that f is convex, and show that measurability cannot be dropped.
2. (Hermite 1883, Hadamard 1893) For convex f on $[a, b]$ prove

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f \leq \frac{f(a)+f(b)}{2},$$

and describe the equality cases.

3. (classical) Let f be convex with $f(0) = 0$. Prove that $x \mapsto f(x)/x$ is nondecreasing on $(0, \infty)$, deduce that f is subadditive on $[0, \infty)$ when $f(0) = 0$, and show that convexity alone does not imply subadditivity without $f(0) = 0$.
4. (Карачёв 1.113) Let f be analytic in the strip $a < \operatorname{Re} z < b$ and continuous up to the boundary. Prove that $M(x) = \sup_{\operatorname{Re} z=x} |f(z)|$ is logarithmically convex on $[a, b]$ (the Hadamard three-lines theorem), and state the analogous statement for an annulus.
5. (Proposed problems, №462) If $f: [0, 1] \rightarrow \mathbb{R}$ is convex with $f(0) = 0$, prove

$$\frac{1}{6} \left(\int_{1/2}^1 f - \int_0^{1/2} f \right) \geq \int_0^{1/2} x f(x) dx.$$

6. (RMO) Let $a > 0$ and $F = \{f: [0, 1] \rightarrow \mathbb{R} \mid f \text{ concave, } f(0) = 1\}$. Determine $\min_{f \in F} \left(\int_0^1 f - (a+1) \int_0^1 x f \right)$.

7. (RMO) (a) If $f: [0, \infty) \rightarrow [0, \infty)$ is differentiable, convex and $f(x) \leq x$ for all x , prove $f'(x) \leq 1$. (b) Determine all such f with $f(0) = 0$ and $f'(x)f(f(x)) = x$ for all $x \geq 0$.
8. (RMO) Let $f: [0, \infty) \rightarrow [0, \infty)$ be convex and differentiable with $x^s f'(x) \geq (f(f(x)))^s$ for some $s > 1$ and all x . Prove $f(x) \leq x$, and then that the same holds with s replaced by any $t \geq 1$.

9. (Капачёв 1.34) Prove that a continuous function on an interval is convex if and only if at every point

$$\lim_{h \rightarrow +0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} \geq 0.$$

Then prove that $\partial f(x) \neq \emptyset$ at interior points, and that differentiability is equivalent to $\partial f(x)$ being a singleton.

10. (MIPT 2017) Let $u: \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable and strictly convex. Prove that the boundary value problem $\ddot{x} = \nabla u(x)$, $x(0) = a$, $x(1) = b$ has at most one solution.
11. (classical) Prove that a convex function on an interval has at most countably many points of non-differentiability, and construct a convex function whose set of «kinks» is dense.
12. (Капачёв 3.59–3.60) Let $\lambda_1 \geq \dots \geq \lambda_n$ be reals. Prove that numbers $a_{11} \geq \dots \geq a_{nn}$ occur on the diagonal of a symmetric matrix with spectrum (λ_i) if and only if (λ_i) majorises (a_{ii}) (the Schur–Horn theorem), and explain where convexity enters.